HUJI · 769001 · DYNAMICS AND CODING IN THE BRAIN

Week 03

Attractors, Competition and Oscillations

attractor · bistability · memory · mutual inhibition · Jacobian · network modes · pitchfork · symmetry breaking · basin of attraction · decision making · E–I network · Hopf · limit cycle

What this week contains. Part A is Lecture 3, written up from the lecture and the course book [LMLF §2.3.1–2.3.3]. It runs from attractors and bistability of the autapse (where Week 2 stopped), through two neurons that inhibit each other, to an excitatory–inhibitory pair that oscillates. The lecture stopped just before §2.3.4 (all-to-all networks). There was no bonus-hour or TA material this week. Practice problems with full solutions and a cheat sheet close the week. Errors and typos in the book are corrected in pitfall boxes.

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PART A · LECTURE 3 (BOOK §2.3.1-2.3.3)

Attractors, Competition and Oscillations

SECTION 01

Where we are

Week 2 studied the autapse: one neuron with a synapse onto itself, of strength J [LMLF §2.3.1]:

$$\tau \dot{r} = -r + g(Jr + h^{\text{ext}}), \qquad g(x) = \frac{1}{1 + e^{-\beta(x-\theta)}}.$$

What we know about it:

- A **fixed point** r^* solves $r^* = g(Jr^* + h^{\text{ext}})$: it is where the S-curve $g(Jr + h^{\text{ext}})$ crosses the identity line.
- Its **stability** is set by $\lambda = Jg'(Jr^* + h^{\text{ext}})$, the slope of the S-curve at the crossing: $\lambda < 1$ stable, $\lambda > 1$ unstable.
- With the book's values $\theta = 0$, $\beta = 20$, $J = 1$: for $-0.803 < h^{\text{ext}} < -0.197$ there are three fixed points, otherwise one. At the two edges a stable and an unstable fixed point merge and vanish: a **saddle-node bifurcation**.

THE PLAN FOR THIS WEEK: THREE TINY NETWORKS, THREE BEHAVIOURS

1. **One excitatory neuron** (the autapse): two stable states, so it can remember a bit.
2. **Two neurons that inhibit each other**: one wins and silences the other, so they can make a decision.
3. **One excitatory and one inhibitory neuron**: neither settles down, so they oscillate.

The tools are the same every time: find the fixed points, linearize around them, read stability from the eigenvalues. In two dimensions the single number λ becomes a 2×2 matrix, and Week 1's eigenvalue tools take over.

SECTION 02

Attractors and bistability

> 2.1 Attractors and bistability

DEF · STATE SPACE AND ATTRACTOR

The state space is the set of all possible states of the system; for the autapse it is just the r axis. An attractor is a region of state space to which all nearby trajectories converge [LMLF 2.3.1]. A stable fixed point is the simplest attractor; this week we will meet another kind, the limit cycle (§13).

DEF · BISTABILITY

A system is bistable if it has two stable fixed points (two attractors). Between them there is always an unstable fixed point, which separates the starting values that go to one from those that go to the other. The autapse is bistable for $-0.803 < h^{\text{ext}} < -0.197$ (with $\theta = 0$, $\beta = 20$, $J = 1$).

EXAMPLE · WHERE DOES THE NEURON END UP? ($h^{\text{ext}} = -0.5$, $\tau = 10\,\text{MS}$)

t (ms)	0	10	20	30	50	70	100
$r(0) = 0.40$	0.40	0.157	0.058	0.021	0.003	0.000	0.000
$r(0) = 0.45$	0.45	0.194	0.072	0.026	0.004	0.001	0.000
$r(0) = 0.55$	0.55	0.806	0.928	0.974	0.996	0.999	1.000
$r(0) = 0.60$	0.60	0.843	0.942	0.979	0.997	1.000	1.000

(Numerical solutions.) Starting values only 0.1 apart end in opposite states. The final state is decided entirely by which side of 0.5 the neuron starts on: 0.5 is the unstable fixed point.

PITFALL · THREE INTERSECTIONS, NOT TWO

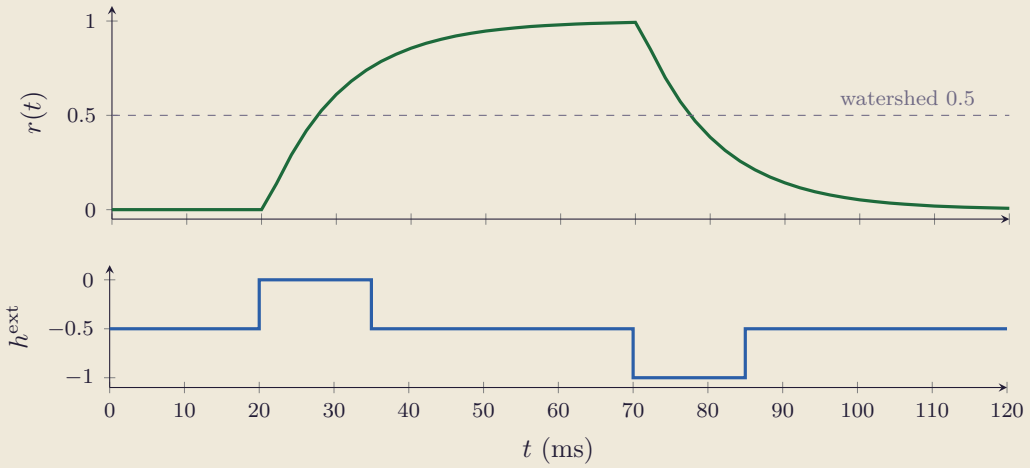
The book says that the bistable regime has "two intersection points". It has *three* intersections (fixed points), of which *two* are stable. The unstable middle one is always there, between the two attractors.

> 2.2 Memory: switching with a brief input

Why care about two stable states? Because a bistable neuron can store one bit [LMLF 2.3.1]. Left alone in the bistable range, it stays forever in whichever state it is in. To write a bit, change h^{ext} *briefly*:

1. The neuron rests in the low state, $h^{\text{ext}} = -0.5$.
2. A brief pulse raises h^{ext} to 0, *above* the bifurcation point -0.197. Now the low fixed point no longer exists; the only attractor is the high one, and r starts climbing.
3. The pulse ends: h^{ext} returns to -0.5. The low state exists again, but if r has already passed the watershed 0.5, it keeps going to the high state.
4. The neuron stays high indefinitely, although the input that put it there is long gone. That is memory.
5. To erase, pulse h^{ext} *below* -0.803 (here to -1): the high state disappears and the neuron falls

back to the low one, where it stays.



State transitions, simulated after [LMLF Fig. 2.9] ($\theta = 0$, $\beta = 20$, $J = 1$, $\tau = 10\,\text{ms}$). A 15 ms pulse to $h^{\text{ext}} = 0$ switches the neuron on (it has reached $0.76 > 0.5$ when the pulse ends); a 15 ms pulse to $h^{\text{ext}} = -1$ switches it off (it has fallen to $0.23 < 0.5$). In between, at the same $h^{\text{ext}} = -0.5$, the neuron is high or low depending only on its history.

THE STATE DEPENDS ON HISTORY

At $t = 10\,\text{ms}$ and at $t = 60\,\text{ms}$ the input is the same, $h^{\text{ext}} = -0.5$, but the rate is 0 in one case and 0.98 in the other. The difference is what happened before. This dependence on past input (often called hysteresis) is what lets a single neuron with feedback hold information; the book develops it into a model of long-term memory in Ch. 5.

PITFALL · THE PULSE MUST BE LONG ENOUGH

Raising h^{ext} is not enough by itself; the rate must cross the watershed 0.5 *before* the pulse ends. With the same parameters, a 5 ms pulse to $h^{\text{ext}} = 0$ brings r only to 0.36. When h^{ext} returns to -0.5, the rate is on the low side of 0.5 and decays back to 0: nothing is stored.

> 2.3 The two ingredients: positive feedback and saturation

What makes bistability possible? We need an S-curve that can cross the identity line three times. Two things are required [LMLF 2.3.1]:

- **Positive feedback** that is strong enough: in the middle crossing the S-curve must be steeper than 1. Its maximal slope is $\beta J/4$ (the effective gain over 4), so we need

$$\boxed{\frac{\beta J}{4} > 1}.$$

  If this fails, the S-curve is flatter than the identity line everywhere and crosses it only once, whatever h^{ext} is. If it holds, a suitable h^{ext} produces three crossings.

- **Saturation**: the curve must bend back below the identity line at high r (and stay above it at low r). That is what the bounded logistic function provides, and what the linear g lacked: there, $J > 1$ meant an unstable fixed point and divergence, with no second stable state to land in.

EXAMPLE · HOW MUCH FEEDBACK IS ENOUGH? ($\beta = 20$)

$\beta J/4 > 1 \iff J > 4/\beta = 0.2$. With the book's $J = 1$: $\beta J/4 = 5$, comfortably bistable for some range of h^{ext}. With $J = 0.1$: $\beta J/4 = 0.5$, one fixed point for every h^{ext}, no memory.

Keep these two ingredients in mind: we will meet them again in the next network, where the positive feedback comes, surprisingly, from inhibition (§7.3).

> 2.4 A technical aside: the case $\lambda = 1$

The linearization decides stability whenever $\lambda \neq 1$. When $\lambda = 1$ exactly, it gives $\tau\dot{\delta r} = 0$, which says nothing. This happens precisely at the bifurcation points, where the S-curve is tangent to the identity line.

Step 1: expand one order further [LMLF eq. 2.23-2.25]. A second-order Taylor expansion adds the curvature term:

$$g(h^* + J\delta r) \approx g(h^*) + g'(h^*)J\delta r + \tfrac{1}{2}g''(h^*)J^2\delta r^2.$$

Step 2: substitute. The constant term vanishes (fixed point) and the linear term vanishes ($\lambda = 1$), leaving

$$\tau\,\dot{\delta r} = \tfrac{1}{2}J^2 g''(h^*)\,\delta r^2.$$

Step 3: read the sign. $\delta r^2 > 0$ on *both* sides of r^*, so $\dot{\delta r}$ has the same sign (that of g'') on both sides. The flow goes towards r^* on one side and away on the other.

DEF · HALF-STABLE FIXED POINT

A fixed point is half-stable if the dynamics approach it from one side and leave it on the other.

Step 4: solve (the equation is separable): $\dfrac{d\delta r}{\delta r^2} = c\,dt$ with $c = \frac{J^2 g''(h^*)}{2\tau}$, so $-\frac{1}{\delta r} = ct + \text{const}$ and

$$\delta r(t) = \frac{1}{1/\delta r(0) - c\,t}.$$

This is hyperbolic, not exponential: much slower near the fixed point.

EXAMPLE · AT THE BIFURCATION POINT $h^{\text{ext}} = -0.803$ ($\beta = 20$, $J = 1$, $\tau = 10\,\text{ms}$)

Here $r^* = 0.947$, and $g'' = \beta^2 g(1-g)(1-2g) = 400 \cdot 0.947 \cdot 0.053 \cdot (-0.894) = -17.9$, so $c = -17.9/20 = -0.89$ per ms.

- From above, $\delta r(0) = +0.01$: $\delta r(t) = 1/(100 + 0.89t)$. After 100 ms it is 0.0053: only halved. (An exponential decay with $\tau = 10$ ms would have shrunk it $e^{10} \approx 22\,000$-fold.)
- From below, $\delta r(0) = -0.01$: $\delta r(t) = 1/(-100 + 0.89t)$ grows in size and the approximation breaks down after about 100 ms; the neuron leaves towards the low state.

PITFALL · LINEARIZATION TELLS YOU ONLY ABOUT THE NEIGHBOURHOOD

Both the first- and second-order expansions are valid only while δr is small. They decide *local* stability. Whether a fixed point is reached from far away (global stability) needs the full picture, such as the vector field [LMLF 2.3.1]. And $\lambda = 1$ exactly is a knife-edge: the slightest change of h^{ext} makes $\lambda \neq 1$, and the exponential behaviour takes over again.

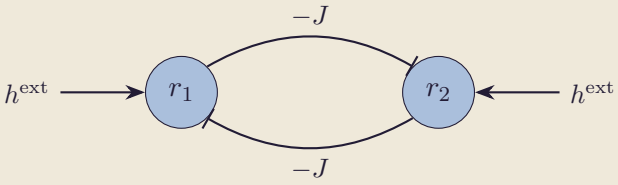
SECTION 03

Mutual inhibition: the model

After one neuron, the next simplest network has two [LMLF §2.3.2].

DEF · MUTUAL INHIBITION

Two neurons (or two populations) are in mutual inhibition if each one inhibits the other. Here the inhibition is symmetric: both synapses have the same strength $J > 0$.



Two neurons inhibiting each other, redrawn after [LMLF Fig. 2.10]. A flat bar at the end of a connection means inhibition. Both neurons get the same external input h^{ext}.

The rate equations, with time measured in units of τ (so $\tau = 1$) and g the logistic function [LMLF eq. 2.29]:

$$\dot{r}_1 = -r_1 + g(-Jr_2 + h^{\text{ext}}), \qquad \dot{r}_2 = -r_2 + g(-Jr_1 + h^{\text{ext}})$$

- Neuron 1 hears neuron 2 through the term $-Jr_2$. Since $J > 0$, the *minus* sign makes the synapse inhibitory: the more neuron 2 fires, the smaller neuron 1's input.
- There are no self-connections: neuron 1's input does not contain r_1.
- Both neurons get the same external input h^{ext}. That makes the two neurons interchangeable, a symmetry that will matter.

PITFALL · WHERE IS THE INHIBITION?

The inhibition is the *sign in front of* J, not the sign of J. The book takes $J > 0$ and writes $-J$ explicitly. Do not also make J negative: $-(-J)r_2$ would be excitation.

SECTION 04

The homogeneous fixed point

> 4.1 The symmetric solution

The fixed points solve $\dot{r}_1 = \dot{r}_2 = 0$:

$$r_1^* = g(-Jr_2^* + h^{\text{ext}}), \qquad r_2^* = g(-Jr_1^* + h^{\text{ext}}).$$

Swap the names $1 \leftrightarrow 2$: the pair of equations is unchanged. So it is natural to look for a solution that is unchanged too, with both neurons firing at the same rate.

DEF · HOMOGENEOUS FIXED POINT

A fixed point is homogeneous if all neurons have the same rate, $r_1^* = r_2^* = r^*$. For mutual inhibition it solves

$$r^* = g(-Jr^* + h^{\text{ext}}) \qquad \text{[LMLF eq. 2.32]}.$$

> 4.2 There is always exactly one

The equation $r^* = g(-Jr^* + h^{\text{ext}})$ has exactly one solution, for every $J > 0$ and every h^{ext}. The argument, step by step:

1. Compare the left side $L(r) = r$ with the right side $R(r) = g(-Jr + h^{\text{ext}})$ on $0 \le r \le 1$ (the logistic g is always between 0 and 1).
2. At $r = 0$: $L = 0$ and $R = g(h^{\text{ext}}) > 0$. So $R > L$.
3. At $r = 1$: $L = 1$ and $R = g(-J + h^{\text{ext}}) < 1$. So $R < L$.
4. L is increasing. R is *decreasing*, because a larger r means more inhibition, hence less input, hence a smaller g.
5. A continuous curve that starts above and ends below must cross at least once; an increasing and a decreasing curve can cross at most once. So: exactly one crossing.

COMPARE WITH THE AUTAPSE

In the excitatory autapse the right side $g(Jr + h^{\text{ext}})$ was *increasing*, an S-curve that could cross the identity line three times. Inhibition flips it into a falling curve, which can cross only once. So the *homogeneous* equation never has more than one solution. If bistability appears, it must come from *non-homogeneous* fixed points, with $r_1^* \neq r_2^*$. That is exactly what will happen.

EXAMPLE · THE BOOK'S FIG. 2.11 PARAMETERS: $\theta = 0$, $\beta = 5$, $J = 2$, $h^{\text{ext}} = 1$

Try $r^* = 0.5$: the input is $-2(0.5) + 1 = 0$, and $g(0) = 0.5$ ✓. So $r^* = 0.5$ exactly, with input $h^* = 0$.

EXAMPLE · THE BOOK'S FIG. 2.12 PARAMETERS: $\theta = 0$, $\beta = 1$, $h^{\text{ext}} = 1$, $J = 3$

No guess works exactly, so iterate or bisect: $r^* = 0.4288$. Check: input $h^* = -3(0.4288) + 1 = -0.286$, and $g(-0.286) = 1/(1 + e^{0.286}) = 0.429$ ✓.

SECTION 05

Linearizing: the Jacobian

> 5.1 Perturbing two neurons

Is the homogeneous fixed point stable? As in Week 2, nudge the state slightly and see whether the nudge shrinks or grows. The new feature: there are now two numbers to nudge, and nudging one affects the other.

Write $r_i(t) = r^* + \delta r_i(t)$ for $i = 1, 2$, and call $h^* = -Jr^* + h^{\text{ext}}$ the input at the fixed point. Step by step for neuron 1:

1. Substitute: $\frac{d}{dt}(r^* + \delta r_1) = -(r^* + \delta r_1) + g(-J(r^* + \delta r_2) + h^{\text{ext}}) = -r^* - \delta r_1 + g(h^* - J\delta r_2)$.
2. r^* is a constant, so the left side is just $\dot{\delta r}_1$.
3. First-order Taylor expansion of g around h^*: $g(h^* - J\delta r_2) \approx g(h^*) + g'(h^*)(-J\delta r_2)$.
4. $-r^* + g(h^*) = 0$, because r^* is a fixed point. What is left:

$$\dot{\delta r}_1 = -\delta r_1 - Jg'(h^*)\,\delta r_2.$$

5. Neuron 2 is the same with $1 \leftrightarrow 2$: $\dot{\delta r}_2 = -\delta r_2 - Jg'(h^*)\,\delta r_1$.

Call $a \equiv Jg'(h^*)$, the effective coupling at the fixed point. In matrix form [LMLF eq. 2.36-2.37]:

$$\dot{\vec{\delta r}} = M\,\vec{\delta r}, \qquad M = \begin{pmatrix} -1 & -a \\ -a & -1 \end{pmatrix}, \qquad a = Jg'(h^*) > 0$$

$a > 0$ because $J > 0$ and the logistic function is increasing ($g' > 0$).

A SHORTCUT FOR g' AT A FIXED POINT

For the logistic function $g' = \beta g(1 - g)$. At the fixed point $g(h^*) = r^*$, so

$$g'(h^*) = \beta r^*(1 - r^*), \qquad a = \beta J\, r^*(1 - r^*).$$

No exponentials needed once r^* is known. Example ($\beta = 1$, $J = 3$, $r^* = 0.4288$): $a = 3 \cdot 0.4288 \cdot 0.5712 = 0.735$.

> 5.2 The Jacobian

DEF · JACOBIAN MATRIX

Write the dynamics as $\dot{\vec{r}} = \vec{F}(\vec{r})$; here $F_1 = -r_1 + g(-Jr_2 + h^{\text{ext}})$ and $F_2 = -r_2 + g(-Jr_1 + h^{\text{ext}})$. The Jacobian at a point $\vec{r}^*$ is the matrix of all first partial derivatives,

$$M_{ij} = \frac{\partial F_i}{\partial r_j}\Big|_{\vec{r}^*}.$$

Row i says how $\dot{r}_i$ responds to a small change of each rate. Near a fixed point the dynamics of the perturbation are $\dot{\vec{\delta r}} = M\vec{\delta r}$: the Jacobian is the many-dimensional version of Week 2's single number $\lambda - 1$. It depends on the point where you evaluate it.

EXAMPLE · THE SAME M FROM PARTIAL DERIVATIVES

$\partial F_1/\partial r_1 = -1$ (the leak), $\partial F_1/\partial r_2 = g'(\cdot) \cdot (-J) = -Jg'(h^*) = -a$ (chain rule). Likewise $\partial F_2/\partial r_1 = -a$ and $\partial F_2/\partial r_2 = -1$. This is the matrix above. ✓

WHAT THE OFF-DIAGONAL ENTRIES DO

The diagonal entries (-1, the leak) describe each neuron on its own; the off-diagonal entries ($-a$) are the coupling. If M were diagonal, each perturbation would evolve alone, $\dot{\delta r}_i = M_{ii}\delta r_i$, and the fixed point would be stable iff every $M_{ii} < 0$. Here M is not diagonal: pushing neuron 1 up pushes neuron 2 down. We cannot judge stability neuron by neuron. We need directions in which the dynamics *do* decouple: the eigenvectors.

SECTION 06

Network modes: eigenvectors of the Jacobian

> 6.1 From neurons to modes

DEF · NETWORK MODE

A mode is a fixed pattern of activity across the neurons, given by an eigenvector $\vec{u}$ of the Jacobian: changing every neuron by the amounts in $\vec{u}$ [LMLF §2.3.2]. A perturbation along an eigenvector stays along it, $\vec{\delta r}(t) = c(t)\vec{u}$ with $\dot{c} = \mu c$, so $c(t) = c(0)e^{\mu t}$. Different modes do *not* affect each other.

So in the *neuron* basis the variables are coupled, while in the *mode* (eigenvector) basis they are not. A general perturbation is a sum of modes, and each mode grows or decays on its own.

STABILITY FROM THE EIGENVALUES

If the eigenvectors of M span the space, every perturbation is a combination of modes, and

- all eigenvalues with **negative real part**: every mode decays, the fixed point is stable;
- at least one eigenvalue with **positive real part**: that mode grows, the fixed point is unstable.

For complex eigenvalues only the real part decides (it sets $|e^{\mu t}| = e^{\mathrm{Re}\,\mu\, t}$); the imaginary part makes the mode rotate. We will need this in the E–I network.

> 6.2 The two modes of mutual inhibition

Finding the eigenvectors of $M = \begin{pmatrix} -1 & -a \\ -a & -1 \end{pmatrix}$ needs no characteristic polynomial. Two shortcuts from Week 2 Part B do it:

1. **Equal row sums.** Both rows add up to $-1-a$. So $M\mathbf{1} = -(1+a)\mathbf{1}$: the vector $\vec{u}_{\parallel} = (1,1)$ is an eigenvector with eigenvalue $-(1+a)$.
2. **Symmetry.** $M = M^T$, so eigenvectors with different eigenvalues are orthogonal. The other one is perpendicular to $(1,1)$: $\vec{u}_{\perp} = (1,-1)$.
3. **Check** by multiplying: $M(1,-1)^T = (-1+a,\ -a+1)^T = -(1-a)(1,-1)^T$. Eigenvalue $-(1-a)$.

MODES OF MUTUAL INHIBITION [LMLF EQ. 2.37]

mode	eigenvector	eigenvalue
homogeneous direction (both neurons together)	$\vec{u}_{\parallel} = (1, 1)$	$\mu_{\parallel} = -(1 + a)$
heterogeneous direction (one up, one down)	$\vec{u}_{\perp} = (1, -1)$	$\mu_{\perp} = -(1 - a)$

> 6.3 The full solution, mode by mode

This is the decoupling of Week 2 Part B, now with $a = Jg'(h^*)$ coming from the network. Define the sum and difference of the perturbations, $\delta r_+ = \delta r_1 + \delta r_2$ and $\delta r_- = \delta r_1 - \delta r_2$. Adding and subtracting the two equations:

$$\dot{\delta r}_+ = -(1 + a)\,\delta r_+, \qquad \dot{\delta r}_- = -(1 - a)\,\delta r_-.$$

Each is a one-variable equation from Week 1:

$$\delta r_+(t) = \delta r_+(0)e^{-(1+a)t}, \qquad \delta r_-(t) = \delta r_-(0)e^{-(1-a)t},$$

and back to the neurons, $\delta r_{1,2} = \frac{1}{2}(\delta r_+ \pm \delta r_-)$:

$$\delta r_1(t) = \tfrac{1}{2}\big[\delta r_+(0)e^{-(1+a)t} + \delta r_-(0)e^{-(1-a)t}\big], \qquad \delta r_2(t) = \tfrac{1}{2}\big[\delta r_+(0)e^{-(1+a)t} - \delta r_-(0)e^{-(1-a)t}\big]$$

EXAMPLE · KICK ONE NEURON: $\beta = 1$, $h^{\text{ext}} = 1$, $J = 3$, SO $a = 0.735$

Start with $\delta r(0) = (0.02, 0)$: only neuron 1 is pushed up. Then $\delta r_+(0) = \delta r_-(0) = 0.02$, and $\mu_{\parallel} = -1.735$, $\mu_{\perp} = -0.265$.

t	δr_+	δr_-	δr_1	δr_2
0	0.0200	0.0200	0.0200	0
1	0.0035	0.0153	0.0094	−0.0059
2	0.0006	0.0118	0.0062	−0.0056
4	0.0000	0.0069	0.0035	−0.0035

Read the table:

- Neuron 2 was not touched, yet it goes *down*: neuron 1 is now firing more and inhibits it.
- The sum δr_+ is gone by $t \approx 2$; what remains is a pure difference, $\delta r_1 \approx -\delta r_2$.
- The difference decays too, but slowly (rate 0.265). Both modes decay: the fixed point is stable.

SECTION 07

When the symmetric state becomes unstable

> 7.1 The stability verdict

- $\mu_{\parallel} = -(1 + a) < 0$ **always**, since $a > 0$. The homogeneous fixed point is always stable against perturbations that move both neurons equally.

- $\mu_\perp = -(1-a) < 0$ **iff** $a < 1$. If the inhibition is strong enough that $a > 1$, the heterogeneous mode grows: the homogeneous fixed point is unstable, and the direction of instability is $(1, -1)$.

STABILITY OF THE HOMOGENEOUS STATE

$$\text{stable} \iff a = Jg'(h^*) = \beta J\, r^*(1-r^*) < 1.$$

When it fails, two neurons that start with similar but not equal rates move *apart*.

EXAMPLE · SCANNING J ($\theta = 0$, $\beta = 1$, $h^{\text{ext}} = 1$; THE BOOK'S FIG. 2.12 PARAMETERS)

J	r^*	$a = Jr^*(1-r^*)$	$\mu_\perp = -(1-a)$	homogeneous state
1	0.599	0.240	−0.760	stable
3	0.429	0.735	−0.265	stable
4	0.376	0.939	−0.061	stable (barely)
4.33	0.362	1.000	0	critical
5	0.336	1.116	+0.116	unstable
10	0.224	1.739	+0.739	unstable

As J grows, r^* falls (more inhibition), but a still grows, and crosses 1 at $J_c \approx 4.33$.

PITFALL · a IS NOT PROPORTIONAL TO J

$a = Jg'(h^*)$ depends on J twice: directly, and through r^* (hence h^* and the slope g'). So the critical J cannot be read off as "$J = 1/g'$". It has to be found by solving $r^* = g(-Jr^* + h^{\text{ext}})$ and $Jr^*(1-r^*) = 1$ together; here numerically, $J_c \approx 4.33$, $r^* \approx 0.362$. A quick bound: $r(1-r) \le \frac{1}{4}$, so with $\beta = 1$, $a \le J/4$, and the homogeneous state is stable for every $J < 4$, whatever h^{ext} is.

EXAMPLE · THE FIG. 2.11 PARAMETERS ARE ALREADY UNSTABLE

With $\beta = 5$, $J = 2$, $h^{\text{ext}} = 1$ we found $r^* = 0.5$. Then $a = \beta J r^*(1-r^*) = 5 \cdot 2 \cdot 0.25 = 2.5 > 1$. The homogeneous state there is unstable: the figure shows a fixed point that the network will not stay at.

The input matters too: a depends on the slope $g'(h^*)$, and h^* depends on h^{ext}. So changing h^{ext} alone can also destabilize the homogeneous state [LMLF §2.3.2].

> 7.2 Symmetry breaking and the pitchfork

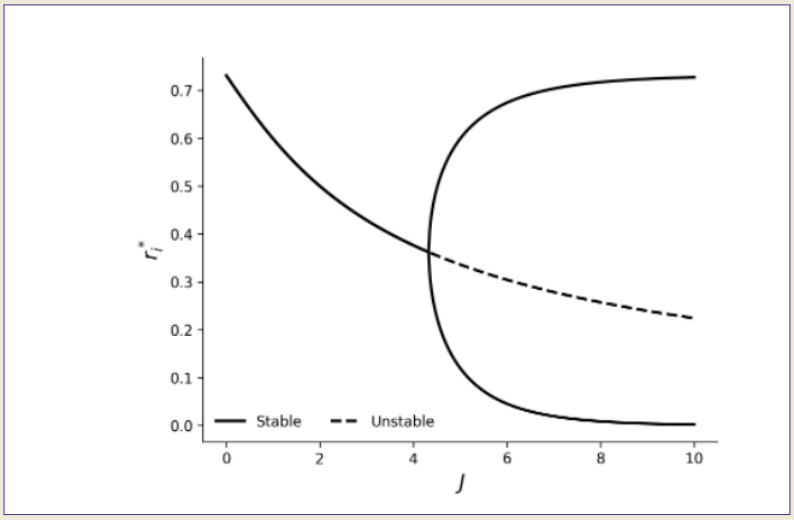
What does the network do once the homogeneous state is unstable? Because the connectivity is symmetric, the only attractors are fixed points (the book proves this in §2.4 with an energy function; here we take it as given). New fixed points must appear, and they do.

DEF · SYMMETRY BREAKING

The equations are unchanged when the two neurons are swapped, but a solution need not be. A fixed point with $r_1^* \neq r_2^*$ breaks the symmetry. The symmetry survives only in the *set* of solutions: if (r_A, r_B) is a fixed point, so is (r_B, r_A). They come in mirror pairs.

DEF · PITCHFORK BIFURCATION

A pitchfork bifurcation is a bifurcation in which one stable fixed point becomes unstable and, at the same moment, two new stable fixed points branch off from it, symmetric to each other. In the bifurcation diagram one line splits into three, like the prongs of a fork. Compare Week 2: in a *saddle-node* bifurcation two fixed points merge and vanish; here none vanish, one changes stability and two are born. (The tanh population of TA session 2 had a pitchfork too, at $J = 1$.)



[LMLF Fig. 2.12]. The fixed-point rates r_i^* against J ($h^{\text{ext}} = 1$, $\beta = 1$, $\theta = 0$). Solid: stable; dashed: unstable. Left of $J_c \approx 4.33$: one homogeneous fixed point, stable. Right of it: the homogeneous one continues as the dashed line (unstable), and two stable asymmetric fixed points appear. On each of them one neuron is high (upper branch) and the other low (lower branch).

EXAMPLE · THE ASYMMETRIC FIXED POINTS AT $J = 5$

Numerically, $(r_1^*, r_2^*) = (0.599,\ 0.120)$, and its mirror image $(0.120,\ 0.599)$. Check the first:

- neuron 1: input $-5(0.120) + 1 = 0.400$, $g(0.400) = 1/(1 + e^{-0.4}) = 0.599$ ✓;
- neuron 2: input $-5(0.599) + 1 = -1.995$, $g(-1.995) = 1/(1 + e^{1.995}) = 0.120$ ✓.

At $J = 10$ the asymmetry is stronger: $(0.727,\ 0.0019)$. The loser is practically silent.

EXAMPLE · WATCHING ONE NEURON WIN ($J = 5$, START $r(0) = (0.35,\ 0.33)$)

The start is close to the unstable homogeneous point $(0.336, 0.336)$, with neuron 1 slightly ahead. Simulating:

t (units of τ)	0	5	10	15	20	25	30	40
r_1	0.350	0.354	0.368	0.392	0.432	0.485	0.538	0.588
r_2	0.330	0.319	0.305	0.282	0.247	0.203	0.163	0.127

The small head start of 0.02 grows (at first like $e^{+0.116t}$, the rate $\mu_\perp$) until saturation takes over and the state settles at $(0.599, 0.120)$. Starting from $(0.33, 0.35)$ instead gives the mirror image: neuron 2 wins.

>7.3 Where does the positive feedback come from? Disinhibition

In §2.3 we said bistability needs positive feedback and saturation. The saturation is the logistic g again. But every synapse here is *inhibitory*: where is the positive feedback?

DEF · DISINHIBITION

Disinhibition is the removal of inhibition: inhibiting an inhibitory neuron reduces the inhibition it sends out, which *excites* its targets indirectly.

Follow one loop, step by step [LMLF §2.3.2]:

1. Neuron 1 fires a bit more.
2. So it inhibits neuron 2 more: neuron 2 fires less.
3. So neuron 2 inhibits neuron 1 less.
4. So neuron 1 fires even more. Back to step 1.

Two minus signs make a plus: around the loop, neuron 1 excites itself through neuron 2. That loop is the heterogeneous mode, and its gain is a: when $a > 1$ it beats the leak and the difference grows, exactly like an autapse with $\lambda > 1$.

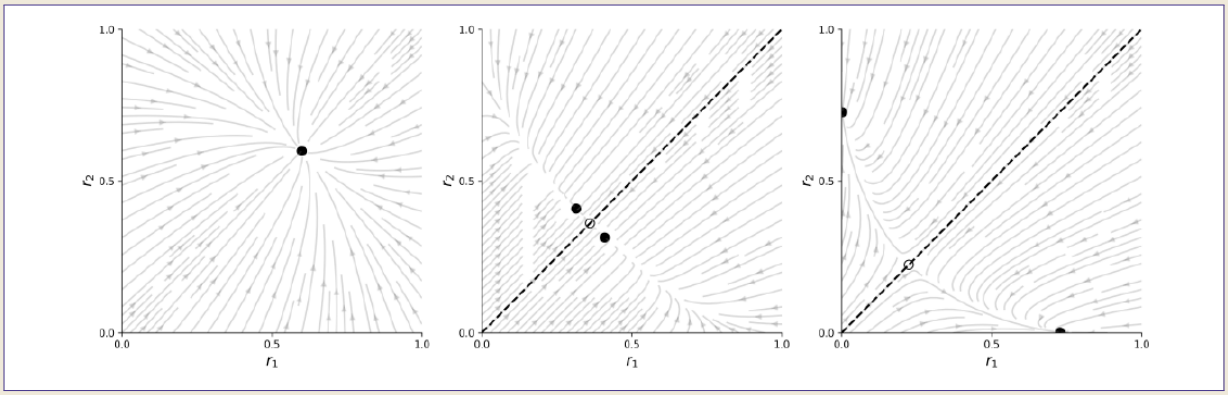
SAME RECIPE AS THE AUTAPSE

	excitatory autapse	**mutual inhibition**
positive feedback	the self-synapse J	disinhibition loop, gain a
saturation	logistic g	logistic g
instability when	$\lambda = Jg' > 1$	$a = Jg' > 1$
two attractors	low / high rate	neuron 1 wins / neuron 2 wins
bifurcation	saddle-node (vary h^{ext})	pitchfork (vary J)

SECTION 08

Phase portrait, basins and the separatrix

With two neurons the state is a point (r_1, r_2) in the plane, and we can draw the whole flow at once: a **phase portrait** (Week 1).



[LMLF Fig. 2.13]. Trajectories (grey) in the (r_1, r_2) plane for $J = 1$, 4.35 and 10 (left to right), with $\theta = 0$, $\beta = 1$, $h^{\text{ext}} = 1$. Full circles: stable fixed points; empty circles: unstable. Dashed: the separatrix $r_1 = r_2$. Left: everything flows to the homogeneous point $(0.60, 0.60)$. Middle (just past J_c): the two winners sit close to the diagonal. Right: the winners are near the axes, $(0.73, 0.002)$ and $(0.002, 0.73)$.

PITFALL · TYPO IN THE BOOK'S CAPTION

The caption of [LMLF Fig. 2.13] reads "$J = 1, 435, 10$". It means $J = 1$, 4.35, 10. The middle value is just above the critical $J_c \approx 4.33$, which is why its two stable fixed points are still close to the unstable homogeneous one ($r_1^* \approx 0.31$ and 0.41, around 0.36).

DEF · BASIN OF ATTRACTION

The basin of attraction of an attractor is the set of all initial states whose trajectories converge to it.

DEF · SEPARATRIX

A separatrix is a curve in state space that separates two basins of attraction. A trajectory that starts exactly on it never reaches either attractor (here it goes to the unstable fixed point).

For mutual inhibition with equal inputs and $J > J_c$:

- basin of $(r_1^* > r_2^*)$: all starts with $r_1 > r_2$;
- basin of $(r_1^* < r_2^*)$: all starts with $r_1 < r_2$;
- separatrix: the diagonal $r_1 = r_2$. A start on it keeps $r_1 = r_2$ forever (by symmetry) and slides to the unstable homogeneous point.

Whoever starts ahead wins, however small the lead. This is the two-dimensional version of the autapse's watershed at $r = 0.5$.

PITFALL · THE DIAGONAL IS THE SEPARATRIX ONLY BECAUSE OF THE SYMMETRY

That the separatrix is exactly $r_1 = r_2$ relies on the two neurons being identical and getting identical input. With different inputs $h_1^{\text{ext}} \neq h_2^{\text{ext}}$ the symmetry is gone, and the boundary between the basins moves: the neuron with the stronger input can win even from a small deficit.

SECTION 09

Decision making and perceptual bistability

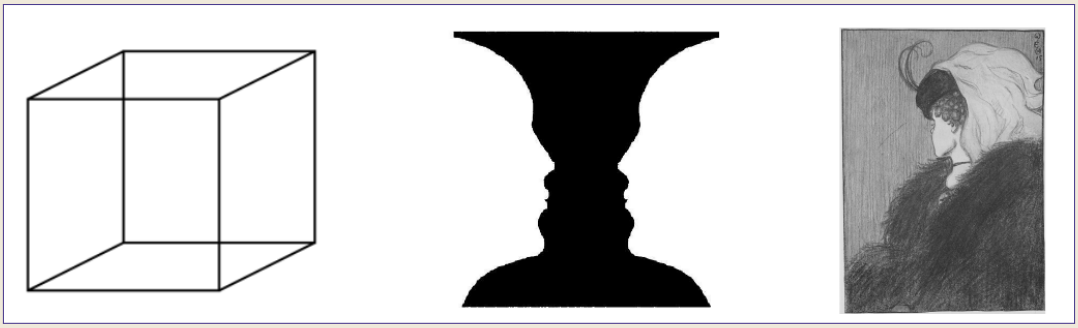
The book's interpretation [LMLF §2.3.2]: let r_1 and r_2 be two *populations* of neurons, each standing for one choice.

- The external inputs $h_1^{\text{ext}}(t)$ and $h_2^{\text{ext}}(t)$ are the **evidence** for each option, arriving over time.
- The initial state is a prior **bias** towards one option.
- Mutual inhibition makes the populations compete. The network ends in one of the two asymmetric attractors: the decision. Ending at $r_1^* > r_2^*$ means "do action 1".

Mutual inhibition between populations is considered a neural basis of binary decision making; the book returns to it in §4.2.2.

DEF · PERCEPTUAL BISTABILITY

Perceptual bistability: an ambiguous image has two interpretations, but at any moment we perceive only one of them, never a blend.



[LMLF Fig. 2.14]. Left: the Necker cube (which face is in front?). Middle: Rubin's vase (a vase, or two faces in profile?). Right: "My Wife and My Mother-in-Law" (a young or an old woman?).

The mutual-inhibition explanation: one population per percept; the ambiguous image drives both; inhibition lets only one stay active; which one depends on the initial state and on noise.

PITFALL · WHAT THE MODEL DOES NOT EXPLAIN

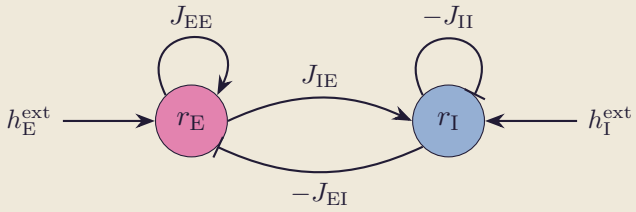
Stare at the Necker cube long enough and the percept *flips* every few seconds. Our model cannot do that: once in an attractor, it stays forever. Explaining the flips needs extra ingredients, such as slow *adaptation* (the winner gets tired) and *noise* [LMLF §2.3.2].

SECTION 10

Excitation and inhibition: the model

So far the networks were purely excitatory (the autapse) or purely inhibitory (mutual inhibition). The minimal network with *both* kinds has one excitatory neuron, E, and one inhibitory neuron, I

[LMLF §2.3.3].



One excitatory and one inhibitory neuron, redrawn after [LMLF Fig. 2.15]. Arrowheads: excitation; bars: inhibition.

The equations ($\tau = 1$) [LMLF eq. 2.38-2.39]:

$$\dot{r}_E = -r_E + g(h_E), \qquad \dot{r}_I = -r_I + g(h_I),$$
$$h_E = J_{EE} r_E - J_{EI} r_I + h_E^{ext}, \qquad h_I = J_{IE} r_E - J_{II} r_I + h_I^{ext}.$$

- $J_{\alpha\beta} > 0$ is the synapse from β to α (second index = source, as in Week 1). So J_{EI} is the inhibition that I sends *to* E.
- Excitatory or inhibitory is again the $\pm$ sign in front: E's synapses enter with $+$, I's with $-$ (Dale's principle, Week 1: a neuron is either excitatory or inhibitory for all its targets).
- Both external inputs are excitatory: $h_E^{ext} > 0$, $h_I^{ext} > 0$.

DEF · THRESHOLD-LINEAR ACTIVATION (RECAP OF WEEK 1)

This time g is the threshold-linear function: $g(h) = h$ for $h \geq 0$ and $g(h) = 0$ for $h < 0$, i.e. $g(h) = \max(h, 0)$. Below threshold the neuron is silent; above it the rate equals the input. There is no saturation from above.

PITFALL · READING J_{EI} BACKWARDS

J_{EI} is *not* "from E to I". The first index is the receiver. In h_E the inhibitory term is $-J_{EI} r_I$: E receives from I. A quick check: every term in h_α has the form $J_{\alpha\beta} r_\beta$, with the source index β appearing twice, next to each other.

SECTION 11

The fixed point of the E–I network

> 11.1 Solving in the linear regime

The model can do several things depending on its six parameters; the book follows only the regime where something new happens: oscillations.

Step 1: drop the threshold. As long as both inputs are positive, $g(h) = h$, and the dynamics are linear [LMLF eq. 2.40]:

$$\dot{\vec{r}} = -\vec{r} + J\vec{r} + \vec{h}^{ext}, \qquad J = \begin{pmatrix} J_{EE} & -J_{EI} \\ J_{IE} & -J_{II} \end{pmatrix}, \quad \vec{r} = \begin{pmatrix} r_E \\ r_I \end{pmatrix}.$$

(Here J is the *connectivity matrix*, with the signs included.)

Step 2: set $\dot{\vec{r}} = 0$. $(I - J)\vec{r}^{*} = \vec{h}^{\text{ext}}$, so

$$\vec{r}^{*} = (I - J)^{-1}\vec{h}^{\text{ext}} \qquad \text{[LMLF eq. 2.41]}.$$

Step 3: invert the 2×2 matrix (swap the diagonal, negate the off-diagonal, divide by the determinant):

$$I - J = \begin{pmatrix} 1 - J_{\text{EE}} & J_{\text{EI}} \\ -J_{\text{IE}} & 1 + J_{\text{II}} \end{pmatrix}, \qquad (I - J)^{-1} = \frac{1}{\det(I - J)} \begin{pmatrix} 1 + J_{\text{II}} & -J_{\text{EI}} \\ J_{\text{IE}} & 1 - J_{\text{EE}} \end{pmatrix},$$

$$\det(I - J) = (1 - J_{\text{EE}})(1 + J_{\text{II}}) + J_{\text{EI}}J_{\text{IE}}.$$

Step 4: the assumption. The book takes the E↔I loop to be strong compared with the self-connections [LMLF eq. 2.43]:

$$J_{\text{EI}}J_{\text{IE}} > (J_{\text{EE}} - 1)(1 + J_{\text{II}}),$$

which is exactly $\det(I - J) > 0$.

Step 5: is the solution allowed? Step 1 assumed positive inputs, i.e. positive rates. Multiplying out,

$$r_{\text{E}}^{*} \propto (1 + J_{\text{II}})h_{\text{E}}^{\text{ext}} - J_{\text{EI}}h_{\text{I}}^{\text{ext}}, \qquad r_{\text{I}}^{*} \propto J_{\text{IE}}h_{\text{E}}^{\text{ext}} + (1 - J_{\text{EE}})h_{\text{I}}^{\text{ext}},$$

so both are positive when

$$h_{\text{E}}^{\text{ext}} > \frac{J_{\text{EI}}}{1 + J_{\text{II}}}\, h_{\text{I}}^{\text{ext}} \qquad \text{and} \qquad h_{\text{E}}^{\text{ext}} > \frac{J_{\text{EE}} - 1}{J_{\text{IE}}}\, h_{\text{I}}^{\text{ext}}.$$

The book's condition: the input to E must be strong enough compared with the input to I.

EXAMPLE · THE BOOK'S FIG. 2.16 PARAMETERS

$J_{\text{EE}} = 3$, $J_{\text{EI}} = 4$, $J_{\text{IE}} = 1$, $J_{\text{II}} = 0$, $h_{\text{E}}^{\text{ext}} = 1$, $h_{\text{I}}^{\text{ext}} = 0.2$.

1. $I - J = \begin{pmatrix} -2 & 4 \\ -1 & 1 \end{pmatrix}$, $\det(I - J) = (-2)(1) - (4)(-1) = 2 > 0$ ✓ (assumption: $4 \cdot 1 = 4 > (3 - 1)(1) = 2$).
2. $(I - J)^{-1} = \frac{1}{2}\begin{pmatrix} 1 & -4 \\ 1 & -2 \end{pmatrix}$.
3. $\vec{r}^{*} = \frac{1}{2}\begin{pmatrix} 1 - 0.8 \\ 1 - 0.4 \end{pmatrix} = \begin{pmatrix} 0.1 \\ 0.3 \end{pmatrix}$.
4. Check the inputs: $h_{\text{E}} = 3(0.1) - 4(0.3) + 1 = 0.1 = r_{\text{E}}^{*}$ ✓, $h_{\text{I}} = 1(0.1) + 0.2 = 0.3 = r_{\text{I}}^{*}$ ✓. Both positive, so $g(h) = h$ was legitimate.

> 11.2 Are there other fixed points?

With threshold-linear g, a neuron at a fixed point is either active ($r^{*} = h^{*} > 0$) or silent ($r^{*} = 0$, $h^{*} \leq 0$). That gives four cases [LMLF §2.3.3]:

1. **Both active.** The solution above; unique, because $I - J$ is invertible.

2. **Both silent.** Then the only input is the external drive, which is positive, so both would fire. Impossible.
3. **E active, I silent.** I's input is $J_{\mathrm{IE}} r_{\mathrm{E}}^{*} + h_{\mathrm{I}}^{\mathrm{ext}} > 0$, so I would fire. Impossible.
4. **E silent, I active.** Then $r_{\mathrm{I}}^{*} = h_{\mathrm{I}}^{\mathrm{ext}}/(1 + J_{\mathrm{II}})$, and E silent requires $h_{\mathrm{E}} = -J_{\mathrm{EI}} r_{\mathrm{I}}^{*} + h_{\mathrm{E}}^{\mathrm{ext}} < 0$, i.e. $h_{\mathrm{E}}^{\mathrm{ext}} < \frac{J_{\mathrm{EI}}}{1+J_{\mathrm{II}}} h_{\mathrm{I}}^{\mathrm{ext}}$. That contradicts Step 5. Impossible.

So there is exactly one fixed point, with both neurons active. (Fig. 2.16 numbers, case 4: it would need $1 < 4 \cdot 0.2 = 0.8$, false.)

SECTION 12

Stability of the E–I fixed point

> 12.1 Trace, determinant and spirals

Near the fixed point both inputs are positive, so the dynamics are exactly the linear ones of Step 1, and the Jacobian is read off directly [LMLF eq. 2.44]:

$$M = -I + J = \begin{pmatrix} -1 + J_{\mathrm{EE}} & -J_{\mathrm{EI}} \\ J_{\mathrm{IE}} & -1 - J_{\mathrm{II}} \end{pmatrix}, \qquad \operatorname{tr} M = J_{\mathrm{EE}} - J_{\mathrm{II}} - 2, \qquad \det M = \det(I - J).$$

($M = -(I - J)$, and for a 2×2 matrix $\det(-A) = (-1)^2 \det A = \det A$.)

Recall Week 1: the eigenvalues satisfy $\mu_1 + \mu_2 = \operatorname{tr} M$ and $\mu_1 \mu_2 = \det M$, and

$$\mu_{1,2} = \frac{\operatorname{tr} M \pm \sqrt{(\operatorname{tr} M)^2 - 4 \det M}}{2}.$$

- If $(\operatorname{tr} M)^2 < 4 \det M$, the eigenvalues are complex conjugates $\alpha \pm i\omega$ with real part $\alpha = \operatorname{tr} M/2$. The trajectories **spiral** (Week 2 Part C): inwards if $\operatorname{tr} M < 0$, outwards if $\operatorname{tr} M > 0$.
- In that case stability is decided by the trace alone:

STABILITY OF THE E-I FIXED POINT (COMPLEX EIGENVALUES)

$$\operatorname{tr} M < 0 \iff J_{\mathrm{EE}} < 2 + J_{\mathrm{II}} : \text{ stable spiral}; \qquad J_{\mathrm{EE}} > 2 + J_{\mathrm{II}} : \text{ unstable spiral.}$$

Self-excitation of E destabilizes; self-inhibition of I stabilizes.

PITFALL · $\det M > 0$ DOES *NOT* MAKE THE EIGENVALUES COMPLEX

The book says that $\det M > 0$ "implies that the two eigenvalues of M are complex conjugate". That is not right. $\det M > 0$ only says the eigenvalues have the same sign (or are a complex pair). They are complex iff $(\operatorname{tr} M)^2 < 4 \det M$, which works out to

$$(J_{\mathrm{EE}} + J_{\mathrm{II}})^2 < 4 J_{\mathrm{EI}} J_{\mathrm{IE}} \qquad \text{i.e.} \qquad J_{\mathrm{EI}} J_{\mathrm{IE}} > \tfrac{1}{4}(J_{\mathrm{EE}} + J_{\mathrm{II}})^2 :$$

the E↔I loop must beat the self-connections by a margin. **Counterexample**: $J_{\mathrm{EE}} = 3$, $J_{\mathrm{II}} = 0$, $J_{\mathrm{IE}} = 1$, $J_{\mathrm{EI}} = 2.1$. The book's assumption holds ($2.1 > 2$), and $\det M = 0.1 > 0$. But $\operatorname{tr} M = 1$ and $(\operatorname{tr} M)^2 - 4 \det M = 0.6 > 0$: the eigenvalues are *real*, 0.887 and 0.113. An unstable *node*: trajectories leave without spiralling. The book's conclusions hold for its Fig. 2.16 example, where $4 J_{\mathrm{EI}} J_{\mathrm{IE}} = 16 > 9 = (J_{\mathrm{EE}} + J_{\mathrm{II}})^2$.

EXAMPLE · TURNING UP J_{EE} ($J_{EI} = 4$, $J_{IE} = 1$, $J_{II} = 0$, $h_E^{ext} = 1$, $h_I^{ext} = 0.2$)

J_{EE}	tr M	det M	eigenvalues	$\vec{r}^*$	behaviour
1.5	-0.5	3.5	$-0.25 \pm 1.85i$	$(0.057, 0.257)$	stable spiral
2	0	3	$\pm 1.73i$	$(0.067, 0.267)$	borderline
3	$+1$	2	$0.5 \pm 1.32i$	$(0.1, 0.3)$	unstable spiral

At $J_{EE} = 1.5$ a perturbation spirals back in, with damped oscillations of period $2\pi/1.85 \approx 3.4$ (units of τ). At $J_{EE} = 3$ it spirals out.

DEF · HOPF BIFURCATION

A Hopf bifurcation happens when a fixed point loses stability because a pair of complex-conjugate eigenvalues crosses the imaginary axis: the real part goes from negative to positive while the imaginary part stays non-zero. Here that happens at $J_{EE} = 2 + J_{II}$. It is the third bifurcation of the course: saddle-node (fixed points merge and vanish), pitchfork (one becomes unstable and two are born), Hopf (a spiral turns from inward to outward).

> 12.2 Link to TA session 2

Exam 2015 Q2 (Week 2 Part C) was the same idea with tanh instead of threshold-linear g and with the fixed point at $(0, 0)$. There tr $= -2$ was fixed, so the spiral was always stable; oscillations needed $K > J$, the loop beating the self-coupling. The book's model adds what the exam left out: the trace can turn positive, the spiral can turn outward, and then something new must happen.

SECTION 13

Limit cycles: the network oscillates

> 13.1 What happens after the spiral leaves?

When $J_{EE} > 2 + J_{II}$, the only fixed point is unstable, and trajectories spiral away from it. Where do they go?

- If g were linear everywhere, the linear analysis would hold for every $\vec{r}$: the spiral would grow forever and the rates would diverge.
- The threshold stops this. Step by step [LMLF §2.3.3]:
  1. As the spiral grows, I's rate swings high, and E's input $h_E = J_{EE} r_E - J_{EI} r_I + h_E^{ext}$ becomes *negative*.
  2. Then $g(h_E) = 0$: E's rate decays exponentially towards 0. The outward push is switched off.
  3. With E low, I loses its excitatory drive, and its rate falls.
  4. With I low, E's input becomes positive again, and E's rate climbs.
  5. With both inputs positive, the outward spiral resumes. Back to step 1.

Outward push near the fixed point plus inward pull far from it: the trajectories settle on a closed

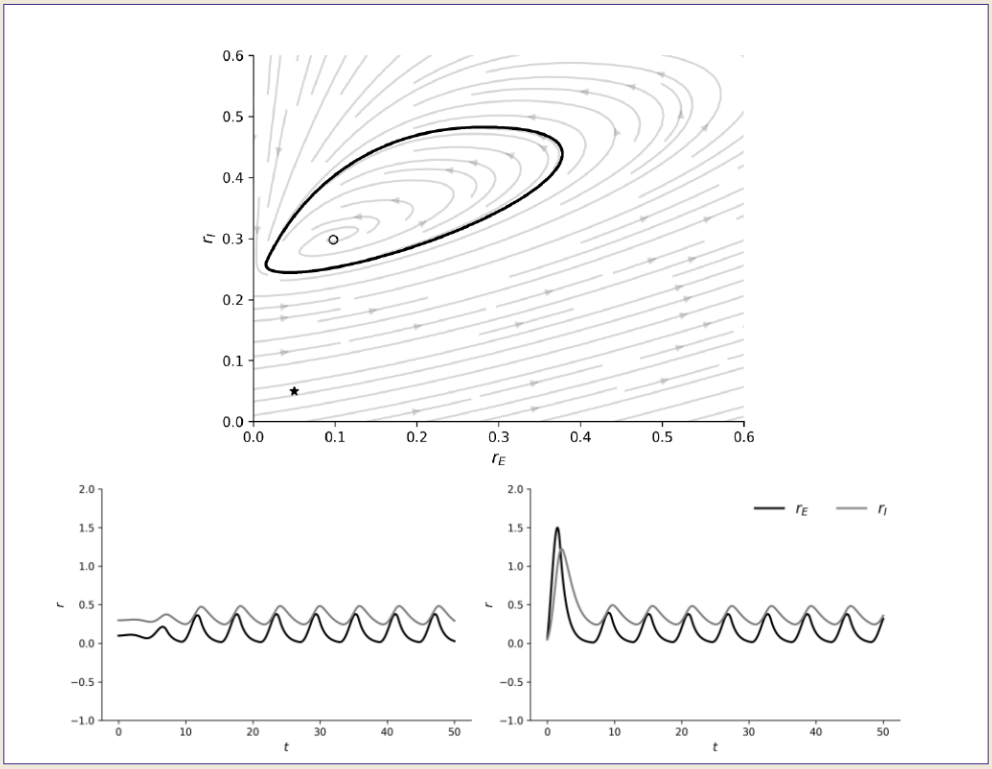
loop in between.

DEF · LIMIT CYCLE

A limit cycle is a closed trajectory that is **isolated**: nearby trajectories are not closed, but spiral towards it (a stable limit cycle) or away from it (unstable). A stable limit cycle is an attractor. Along it the rates repeat periodically: the network **oscillates**.

PITFALL · A CLOSED ORBIT IS NOT AUTOMATICALLY A LIMIT CYCLE

A frictionless pendulum (a harmonic oscillator) also has closed periodic orbits, but a whole family of them: start a bit further out and you stay on a bigger orbit forever. None of them attracts anything, so none is a limit cycle. A limit cycle has a *preferred* amplitude, reached from any start nearby. The borderline case $\operatorname{tr} M = 0$ (purely imaginary eigenvalues, the middle row above) is exactly such a pendulum near the fixed point: a family of closed orbits, none of them attracting.



[LMLF Fig. 2.16]; $J_{\mathrm{EE}} = 3$, $J_{\mathrm{EI}} = 4$, $J_{\mathrm{IE}} = 1$, $J_{\mathrm{II}} = 0$, $h_{\mathrm{E}}^{\mathrm{ext}} = 1$, $h_{\mathrm{I}}^{\mathrm{ext}} = 0.2$. Top: trajectories in the $(r_{\mathrm{E}}, r_{\mathrm{I}})$ plane; thick black curve: the limit cycle; open circle: the unstable fixed point $(0.1, 0.3)$; star: a start far outside, $(0.05, 0.05)$. Bottom: r_{E} (black) and r_{I} (grey) against time, starting near the fixed point (left) and at the star (right). Both end on the same oscillation.

EXAMPLE · THE LIMIT CYCLE IN NUMBERS (FIG. 2.16 PARAMETERS, BY SIMULATION)

- Near the fixed point the linear analysis gives $\mu = 0.5 \pm 1.32i$: the spiral grows by $e^{0.5t}$ and turns with angular frequency 1.32, i.e. period $2\pi/1.32 \approx 4.8$.
- On the limit cycle: r_{E} swings between 0.015 and 0.38, r_{I} between 0.24 and 0.48. The period is ≈ 5.9 (in units of τ; with $\tau = 10\,\mathrm{ms}$ about 59 ms, i.e. roughly 17 Hz).

- Starting near $(0.1, 0.3)$ or at $(0.05, 0.05)$ (where r_E first overshoots to 1.5) gives the *same* final amplitude and period: that is what "attractor" means.
- The period on the cycle (5.9) is longer than the linear one (4.8): part of each cycle is spent with E silent, outside the linear regime.

PITFALL · WHAT WE CANNOT COMPUTE HERE

Finding the limit cycle exactly, and proving it is stable, is much harder than finding fixed points; the book leaves it out. (The method it sketches: in each region where the signs of h_E, h_I are fixed the equations are linear and solvable; glue the pieces so that the end state in one region is the start state in the next.) Here we rely on simulation.

> 13.2 Why it oscillates: delayed negative feedback

Read the cycle in terms of the neurons:

1. E rises ⇒ it drives I up.
2. I high ⇒ it inhibits E, which falls.
3. E low ⇒ I loses drive and falls.
4. I low ⇒ E is released and rises again.

E acts on itself through I with a minus sign: **negative feedback**.

WHY DOES AN INHIBITORY AUTAPSE NOT OSCILLATE?

An inhibitory autapse is negative feedback too. It is just the homogeneous mode of mutual inhibition (§7.3), with eigenvalue $\mu_{\|} = -(1 + a) < 0$: always stable, never oscillating. The difference is delay. In the autapse, the neuron feels its own inhibition at once. In the E–I loop, a change of E reaches E again only after I has had time to respond. By then E has already moved on, so the correction arrives late and overshoots.

EXAMPLE · THE HOTEL SHOWER [LMLF §2.3.3]

You turn on the hot water; it comes out cold, because hot water takes a few seconds to reach the shower head. You turn it up. Still cold. Up more. Now it is pleasant, so you step in, and seconds later it is scalding, because all your earlier turns are arriving. You turn it back, still too hot, back more, and suddenly it is freezing. The rule (hot when cold, cold when hot) is negative feedback; the pipe is the delay. Together they make the temperature **oscillate** around the one you want, an unstable target.

WHY THIS MATTERS IN THE BRAIN

Rhythmic activity is everywhere in the nervous system: circuits that generate walking, swimming, breathing and chewing; rhythms thought to help encode sensory information and to support spatial navigation; and brain rhythms recorded by EEG, used clinically to diagnose epilepsy and sleep disorders. The E–I loop is one of the simplest mechanisms that generates them.

SECTION 14

Summary: three networks

NETWORK	FEEDBACK	INSTABILITY	ATTRACTORS / FUNCTION
Excitatory autapse	positive (self-excitation)	$Jg' > 1$; saddle-node in h^{ext}	two stable rates: **memory**
Mutual inhibition	positive (disinhibition loop)	$a = Jg'(h^*) > 1$; pitchfork in J	one neuron wins: **decision**
E–I pair	delayed negative (E→I→E)	$J_{\text{EE}} > 2 + J_{\text{II}}$; Hopf	limit cycle: **oscillation**

THE METHOD, EVERY TIME

1. Find the fixed points (graphically, by symmetry, or by solving a linear system).
2. Linearize: the Jacobian M.
3. Eigenvalues of M: all real parts negative $\Rightarrow$ stable. For 2×2, use trace and determinant.
4. If a fixed point becomes unstable, ask what the nonlinearity (saturation, threshold) does: new fixed points (bistability) or a limit cycle (oscillation).

WEEK 03

Practice Problems and Summary

SECTION 15

Practice problems with full solutions

EXAMPLE · P1: CAN IT BE BISTABLE AT ALL?

An excitatory autapse with (a) $\beta = 2$, $J = 1$; (b) $\beta = 20$, $J = 0.15$; (c) $\beta = 8$, $J = 1$. For which can a suitable h^{ext} give two stable states?

Solution. Test $\beta J/4 > 1$: (a) 0.5, no. (b) 0.75, no. (c) 2, yes. In (a) and (b) the S-curve is everywhere flatter than the identity line, so it crosses it once whatever h^{ext} is.

EXAMPLE · P2: WRITING A BIT

With $\theta = 0$, $\beta = 20$, $J = 1$, $\tau = 10\,\text{ms}$, the neuron rests at the low state with $h^{\text{ext}} = -0.5$. You can pulse h^{ext} to 0. A 5 ms pulse brings r to 0.36, a 15 ms pulse to 0.76. Which pulse stores a "1"? Why does setting $h^{\text{ext}} = -0.3$ forever not count as memory?

Solution. Only the 15 ms pulse: 0.76 is above the unstable fixed point 0.5 of $h^{\text{ext}} = -0.5$, so the rate continues to the high state; 0.36 is below it and decays back. Memory means the state persists *after* the input returns to its baseline; a permanent change of h^{ext} is just a different input, not a stored bit.

EXAMPLE · P3: HALF-STABLE

At the bifurcation point $h^{\text{ext}} = -0.197$ ($\beta = 20$, $J = 1$), $r^* = 0.053$. Is $g''(h^*)$ positive or negative, and from which side is the fixed point approached?

Solution. $g'' = \beta^2 g(1-g)(1-2g)$ with $g = 0.053 < 0.5$, so $g'' > 0$. Then $\tau\dot{\delta r} = \frac{1}{2}J^2 g''\delta r^2 > 0$ on both sides: r increases. From below ($\delta r < 0$) it moves up towards r^*; from above it moves away, up towards the high state. Half-stable, approached from the left.

EXAMPLE · P4: JACOBIAN AND MODES

Mutual inhibition with $\theta = 0$, $\beta = 1$, $h^{\text{ext}} = 1$, $J = 3$; the homogeneous fixed point is $r^* = 0.4288$. Write the Jacobian, its eigenvectors and eigenvalues, and decide stability.

Solution.

1. $a = \beta J r^*(1 - r^*) = 3 \cdot 0.4288 \cdot 0.5712 = 0.735$.
2. $M = \begin{pmatrix} -1 & -0.735 \\ -0.735 & -1 \end{pmatrix}$.
3. Equal row sums: $(1, 1)$ with $\mu_{\parallel} = -1.735$. Symmetric, so the other is $(1, -1)$ with

$\mu_\perp = -(1 - 0.735) = -0.265$.

4. Both negative: stable. (Check: $\mu_\parallel + \mu_\perp = -2 = \operatorname{tr} M$, $\mu_\parallel \mu_\perp = 0.460 = 1 - a^2 = \det M$ ✓.)

EXAMPLE · P5: FIXED POINT OR NOT?

Same network with $J = 5$. (a) Show that $(0.599, 0.120)$ is a fixed point. (b) The homogeneous fixed point is $r^* = 0.336$. Is it stable?

Solution. (a) $g(-5 \cdot 0.120 + 1) = g(0.4) = 0.599$ and $g(-5 \cdot 0.599 + 1) = g(-1.995) = 0.120$ ✓. (b) $a = 5 \cdot 0.336 \cdot 0.664 = 1.116 > 1$, so $\mu_\perp = +0.116 > 0$: unstable along $(1, -1)$. The asymmetric fixed points are where the network goes.

EXAMPLE · P6: WHICH MODE SURVIVES?

In P4, start at $\vec{\delta r}(0) = (0.02, 0)$. Find δr_1 and δr_2 at $t = 2$. Which pattern remains at late times?

Solution. $\delta r_\pm(0) = 0.02$. At $t = 2$: $\delta r_+ = 0.02e^{-3.47} = 0.0006$, $\delta r_- = 0.02e^{-0.53} = 0.0118$. $\delta r_1 = \frac{1}{2}(0.0006 + 0.0118) = 0.0062$, $\delta r_2 = \frac{1}{2}(0.0006 - 0.0118) = -0.0056$. The homogeneous mode decays 6.5 times faster (1.735/0.265), so at late times the perturbation is a pure difference, $\delta r_1 \approx -\delta r_2$, slowly decaying.

EXAMPLE · P7: NEVER UNSTABLE BELOW $J = 4$

For mutual inhibition with $\beta = 1$, show that the homogeneous fixed point is stable for every $J < 4$, whatever h^{ext} is.

Solution. $a = Jr^*(1 - r^*)$, and $r(1 - r) \le \frac{1}{4}$ for every r (maximum at $r = \frac{1}{2}$). So $a \le J/4 < 1$ when $J < 4$, and $\mu_\perp = -(1 - a) < 0$, $\mu_\parallel < 0$. (With the book's $h^{\text{ext}} = 1$ the actual threshold is $J_c \approx 4.33$, consistent with the bound.)

EXAMPLE · P8: WHO WINS?

Mutual inhibition with $J = 10$ ($\beta = 1$, $h^{\text{ext}} = 1$). The network starts at (a) $(0.20, 0.50)$; (b) $(0.50, 0.49)$; (c) $(0.30, 0.30)$. Where does it end?

Solution. The separatrix is $r_1 = r_2$. (a) $r_2 > r_1$: ends at $(0.0019, 0.727)$. (b) $r_1 > r_2$, however slightly: ends at $(0.727, 0.0019)$. (c) On the separatrix: stays on the diagonal and goes to the unstable homogeneous point $(0.224, 0.224)$; in practice any noise tips it to one side.

EXAMPLE · P9: AN E-I FIXED POINT

$J_{\text{EE}} = 1.5$, $J_{\text{EI}} = 4$, $J_{\text{IE}} = 1$, $J_{\text{II}} = 0$, $h_{\text{E}}^{\text{ext}} = 1$, $h_{\text{I}}^{\text{ext}} = 0.2$. Find $\vec{r}^*$, check it is valid, and classify it.

Solution.

1. $I - J = \begin{pmatrix} -0.5 & 4 \\ -1 & 1 \end{pmatrix}$, $\det = -0.5 + 4 = 3.5$, $(I - J)^{-1} = \frac{1}{3.5}\begin{pmatrix} 1 & -4 \\ 1 & -0.5 \end{pmatrix}$.

2. $\vec{r}^* = \frac{1}{3.5}(1 - 0.8,\ 1 - 0.1) = (0.057,\ 0.257)$. Both positive ✓.

3. $M = \begin{pmatrix} 0.5 & -4 \\ 1 & -1 \end{pmatrix}$: $\operatorname{tr} M = -0.5$, $\det M = 3.5$, $(\operatorname{tr} M)^2 - 4 \det M = 0.25 - 14 < 0$: complex.
4. $\mu = -0.25 \pm 1.85i$: a **stable spiral**. Perturbations die out in damped oscillations with period $2\pi/1.85 \approx 3.4$.

EXAMPLE · P10: SPIRAL OR NODE?

$J_{\mathrm{EE}} = 3$, $J_{\mathrm{EI}} = 2.1$, $J_{\mathrm{IE}} = 1$, $J_{\mathrm{II}} = 0$. Does the fixed point spiral?

Solution. $M = \begin{pmatrix} 2 & -2.1 \\ 1 & -1 \end{pmatrix}$, $\operatorname{tr} M = 1$, $\det M = -2 + 2.1 = 0.1 > 0$. Discriminant $1 - 0.4 = 0.6 > 0$: real eigenvalues $\frac{1}{2}(1 \pm \sqrt{0.6}) = 0.887,\ 0.113$. Both positive: an **unstable node**, no spiralling, although $\det M > 0$. The complex condition $J_{\mathrm{EI}} J_{\mathrm{IE}} > \frac{1}{4}(J_{\mathrm{EE}} + J_{\mathrm{II}})^2$ reads $2.1 > 2.25$: false.

EXAMPLE · P11: WHERE IS THE HOPF POINT?

With $J_{\mathrm{II}} = 0.5$ (other parameters as in Fig. 2.16, eigenvalues complex), for which J_{EE} does the fixed point become unstable?

Solution. $\operatorname{tr} M = J_{\mathrm{EE}} - J_{\mathrm{II}} - 2 = J_{\mathrm{EE}} - 2.5$. Unstable iff $J_{\mathrm{EE}} > 2.5$. Self-inhibition of I moves the Hopf point up by J_{II}. (Check complex at $J_{\mathrm{EE}} = 2.5$: $4 J_{\mathrm{EI}} J_{\mathrm{IE}} = 16 > (2.5 + 0.5)^2 = 9$ ✓.)

EXAMPLE · P12: CONCEPTS

(a) Why can the homogeneous equation of mutual inhibition never have three solutions? (b) Why does the E–I network oscillate but an inhibitory autapse does not? (c) Why does the E–I rate not diverge when the fixed point is unstable?

Solution. (a) r is increasing and $g(-Jr + h^{\mathrm{ext}})$ is decreasing: they cross once. (b) Delay: in the loop, E's change returns through I only after I responds, so the correction overshoots; the autapse's negative feedback is instantaneous and just damps ($\mu_{\|} < 0$). (c) The threshold: when $h_{\mathrm{E}} < 0$ the outward spiral is switched off and E decays to 0, pulling the trajectory back. Push near the fixed point, pull far from it: a stable limit cycle.

SECTION 16

Cheat sheet

BISTABILITY AND MEMORY

- **Attractor**: region all nearby trajectories go to. **Bistable**: two stable fixed points, an unstable one between them as the watershed.
- **Memory**: a brief pulse of h^{ext} past a bifurcation point flips the state, which persists after the pulse (the state depends on history).
- **Ingredients**: positive feedback ($\beta J/4 > 1$) and saturation.

- $\lambda = 1$: second order, $\tau\dot{\delta r} = \frac{1}{2}J^2 g''\delta r^2$: **half-stable**, hyperbolic (slow) approach.

MUTUAL INHIBITION

- $\dot{r}_1 = -r_1 + g(-Jr_2 + h^{\text{ext}})$, $\dot{r}_2 = -r_2 + g(-Jr_1 + h^{\text{ext}})$, $J > 0$.
- Homogeneous fixed point $r^* = g(-Jr^* + h^{\text{ext}})$: always exactly one (rising line vs. falling curve).
- **Jacobian** $M_{ij} = \partial F_i/\partial r_j$: $M = \begin{pmatrix} -1 & -a \\ -a & -1 \end{pmatrix}$, $a = Jg'(h^*) = \beta J r^*(1 - r^*)$.
- Modes: $(1,1)$, $\mu_{\parallel} = -(1+a)$, always stable; $(1,-1)$, $\mu_{\perp} = -(1-a)$, unstable iff $a > 1$.
- $\delta r_{1,2} = \frac{1}{2}[\delta r_+(0)e^{-(1+a)t} \pm \delta r_-(0)e^{-(1-a)t}]$.
- $a > 1$: **pitchfork**, symmetry breaking, two stable winner states. Positive feedback = **disinhibition**.
- Book values $\beta = 1$, $h^{\text{ext}} = 1$: $J_c \approx 4.33$; $J = 5$: $(0.599, 0.120)$; $J = 10$: $(0.727, 0.0019)$.
- Basins $r_1 > r_2$ and $r_1 < r_2$; separatrix $r_1 = r_2$. Decision making; perceptual bistability (flips need adaptation + noise).

EXCITATION AND INHIBITION

- $h_{\text{E}} = J_{\text{EE}}r_{\text{E}} - J_{\text{EI}}r_{\text{I}} + h_{\text{E}}^{\text{ext}}$, $h_{\text{I}} = J_{\text{IE}}r_{\text{E}} - J_{\text{II}}r_{\text{I}} + h_{\text{I}}^{\text{ext}}$; $J_{\alpha\beta}$: from β to α; $g(h) = \max(h, 0)$.
- Fixed point $\vec{r}^* = (I - J)^{-1}\vec{h}^{\text{ext}}$, $\det(I - J) = (1 - J_{\text{EE}})(1 + J_{\text{II}}) + J_{\text{EI}}J_{\text{IE}} > 0$; unique if $h_{\text{E}}^{\text{ext}}$ is large enough relative to $h_{\text{I}}^{\text{ext}}$.
- $M = -I + J$, $\operatorname{tr} M = J_{\text{EE}} - J_{\text{II}} - 2$, $\det M = \det(I - J)$.
- Complex iff $J_{\text{EI}}J_{\text{IE}} > \frac{1}{4}(J_{\text{EE}} + J_{\text{II}})^2$ ($\det > 0$ is *not* enough). Then stable iff $J_{\text{EE}} < 2 + J_{\text{II}}$.
- **Hopf**: complex pair crosses the imaginary axis. Beyond it: stable **limit cycle** (isolated closed orbit, an attractor), kept finite by the threshold.
- Fig. 2.16: $\vec{r}^* = (0.1, 0.3)$, $\mu = 0.5 \pm 1.32i$, cycle period $\approx 5.9\tau$.
- Mechanism: **delayed negative feedback** (the hotel shower).